Storage Lab: Persistent NFS Client Mount & /etc/fstab Configuration

Duration : 30 Min.

Scenario

An application node must mount a shared NFS export (`/srv/nfs/shared`) served by `192.168.10.50` onto local mount directory `/mnt/app_shared`. The mount must persist across server reboots without causing the host to hang during boot if the network is delayed. The administrator must create the mount point, configure a compliant `/etc/fstab` entry with appropriate network mount options, verify the mount and write operations, and record the configuration in `nfs_mount_report.json`.

Task Objectives

Perform the following network filesystem administration tasks:

> **Pre-Staged Files in `student_workspace/`:**
> • **`nfs_mount_report.json`**: Starter NFS mount configuration report with `TODO_*` placeholders.

1. Create the required local mount point directory `/mnt/app_shared` with proper directory ownership and permissions (e.g., `0755`).
2. Configure a persistent mount entry for the remote NFS export `192.168.10.50:/srv/nfs/shared` in `/etc/fstab` with filesystem type `nfs` (or `nfs4`).
3. Apply standard network safety mount options including `_netdev` and `nofail` so the host boots reliably without hanging if network connectivity is delayed.
4. Execute the mount and verify active filesystem reachability and write access inside `/mnt/app_shared`.
5. Record your mount point attributes, fstab configuration line, and verification results in `nfs_mount_report.json`.

Instructions to Perform the Task

1. Open your assessment **Terminal** or storage management CLI interface.
2. Execute the required storage management commands, volume operations, or configuration edits.
3. Update `nfs_mount_report.json` inside `student_workspace/` replacing all `TODO_*`

placeholders with your verified configuration values.

4. Verify your storage configurations locally by executing:

```
python run.py
```

5. Once verified, return to the platform dashboard and click **"Run Tests"** / **"Full Eval"** to evaluate your score!

Validation

Once your storage tasks are executed, return to the platform dashboard and click the **"Run Tests"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Mount point setup: directory /mnt/app_shared exists with proper ownership and permission attributes	3 Marks
TC2	fstab syntax: /etc/fstab contains valid entry with critical network safety options (_netdev, nofail)	4 Marks
TC3	Mount verification: mount command confirms active mount and test file creation succeeds inside /mnt/app_shared	3 Marks

Total Score: 10 Marks

Important Notes

- This is an **Auto-Evaluated** storage assessment.
- Ensure all created volumes, files, and scripts are properly configured and saved before submitting.